gpeyt1-50 · Pile group settlement (equivalent footing)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A pile group with plan dimensions 8 ft × 8 ft (outside-to-outside) carries Q = 900 kips. The friction piles are 45 ft long in a stiff clay crust that overlies a 20 ft normally consolidated clay layer from 45 ft to 65 ft (γsat = 118 pcf, e0 = 1.0, Cc = 0.36). Water table at the ground surface; stiff crust also γsat = 118 pcf. Place the equivalent footing at 2/3 of the embedded length and spread load at 2:1; evaluate one layer at mid-depth of the NC clay.

Most nearly, the consolidation settlement of the group is:

A. 4.5 in.
B. 12.1 in.
C. 1.3 in.
D. 2.2 in.

Given equations / data

> Equivalent footing at D′ = (2/3)D; Δσ = Q/[(B + z)(L + z)] with z measured from D′; S = Cc H/(1 + e0) log(σ′f/σ′0)

Worked solution

1. D′ = (2/3)(45) = 30 ft; mid-clay depth = 55 ft → z = 25 ft
2. Δσ = 900,000/(8+25)² = 826 psf
3. σ′0 = 55(118−62.4) = 3,058 psf
4. S = 0.36×20/2.0 × log(3,884/3,058) = 4.5 in.

Final answer

4.5 in.

Why the other values are wrong

- **12.1 in.** — placed the equivalent footing at the pile toe
- **1.3 in.** — spread the load from the ground surface
- **2.2 in.** — used total stress for σ′0

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7 equivalent-footing method (2/3 D)
- NCEES PE Civil Reference Handbook v2.1 §3.5.1 2:1 method; §3.2.1

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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